



Mission Planning and Trajectory Design of Roundtrip Mars Transfers

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What is a Mars Architecture?

To define the Mars Architecture, we must answer these six key questions.

Why
We Will Go

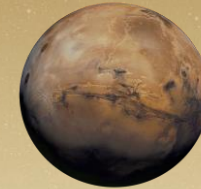
Where
We Will Go

How
We Will Get
There and
Back

When
We Will Go

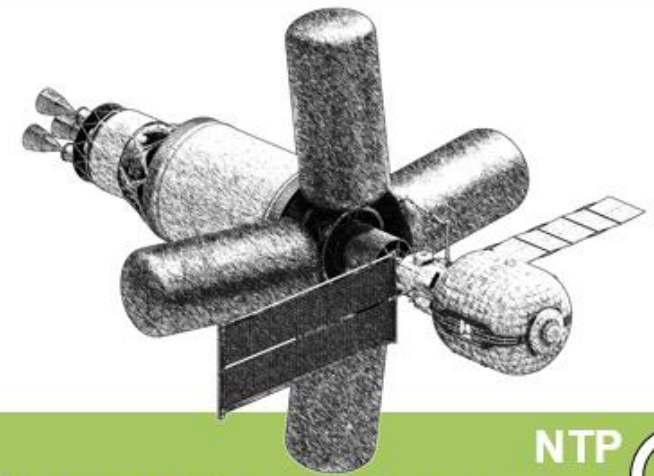
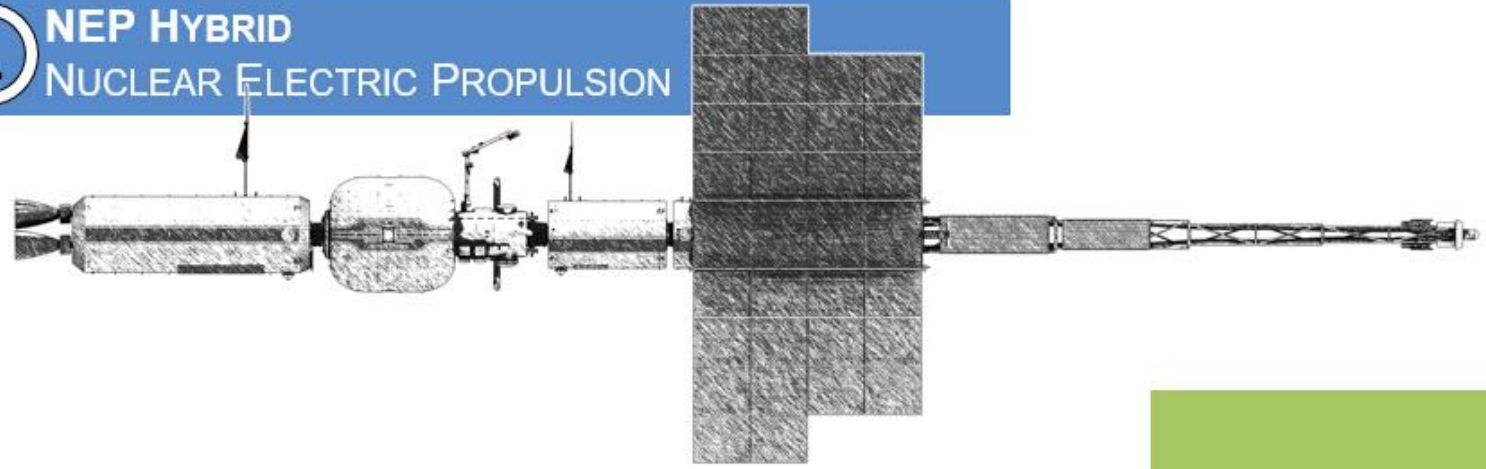
Who
Will Be
Involved

What
We Will Do
There



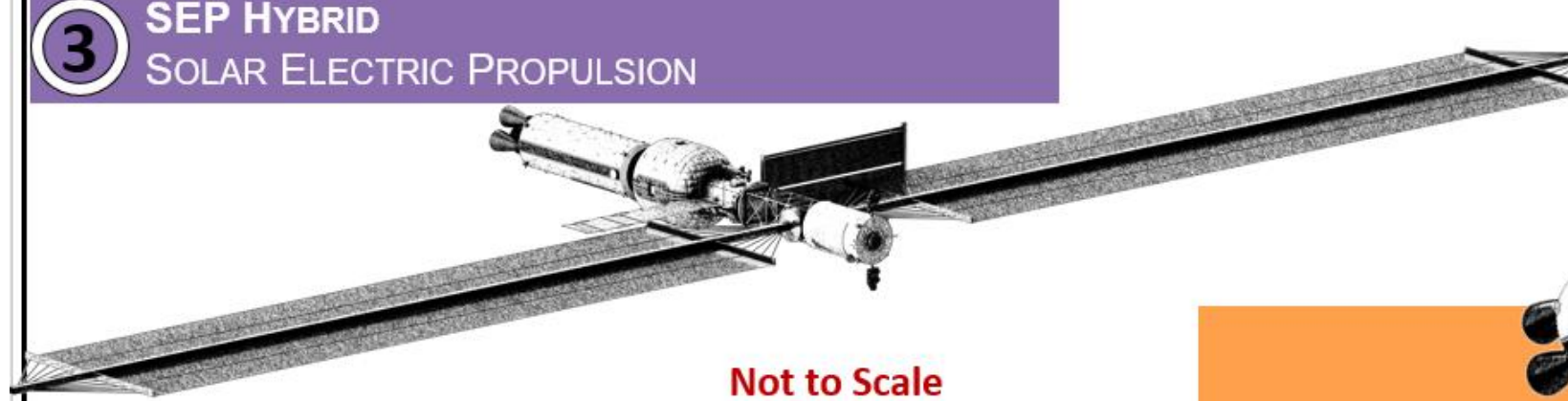
Mars Transportation Options

1 NEP HYBRID NUCLEAR ELECTRIC PROPULSION



NTP 2 NUCLEAR THERMAL PROPULSION

3 SEP HYBRID SOLAR ELECTRIC PROPULSION



ALL-CHEM 4 CHEMICAL PROPULSION

Not to Scale
Artistic Conceptual Elements Shown

1&3 Previously Published in AAS 16-255

2&4 are the focus of this presentation

Modeling and Analyzing High Thrust End-to-End Trajectories

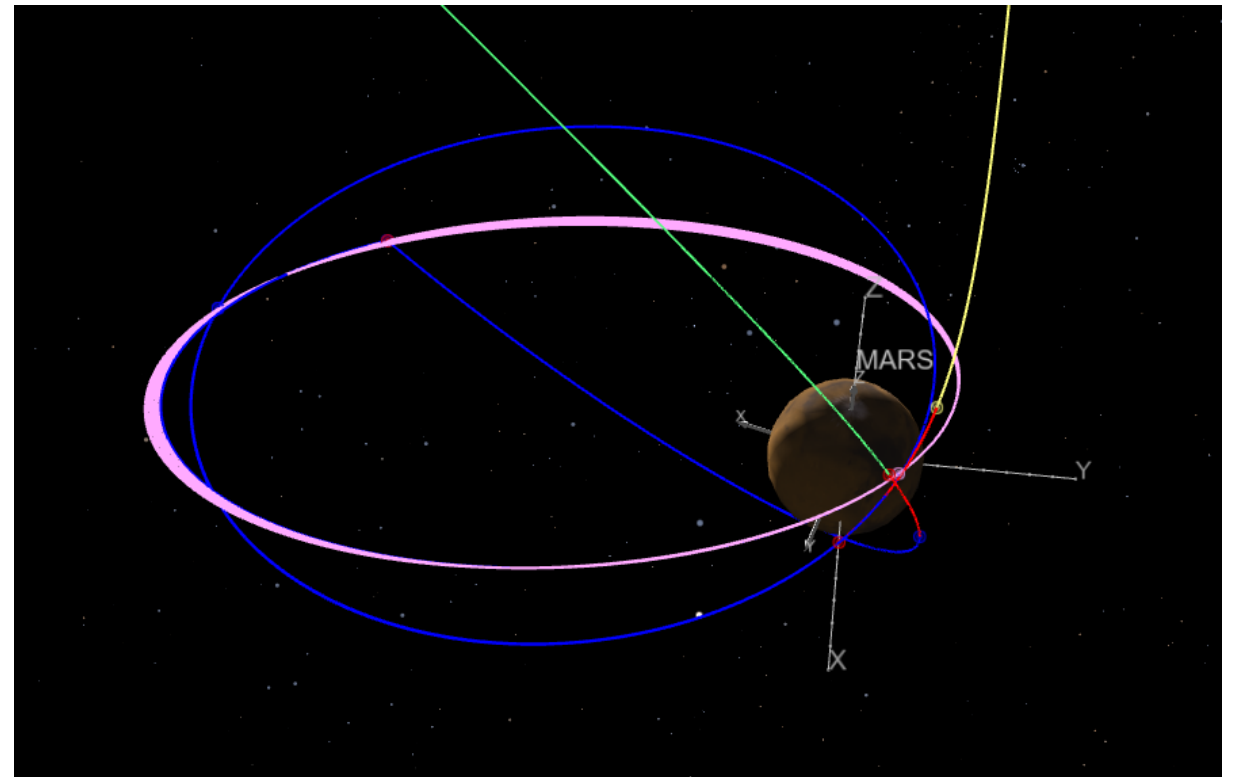


- Heliocentric Transfers
 - Lambert problem
 - Perform scans to find departure/arrival opportunities
- Mars Vicinity Optimization
 - Orbit reorientation
 - Mars parking orbit targeting desired landing sites
 - Bi-elliptic apotwist maneuver sequence to connect incoming/outgoing V_∞ vectors
- Globally Optimized Solution
 - Optimize heliocentric and Mars vicinity trajectories simultaneously
 - Using SNOPT (Sparse Nonlinear OPTimizer) to search for optimal solutions

Bi-Elliptic Apotwist Maneuvers



- 7-Maneuver sequence, insert and depart from high elliptic parking orbits
 - 3-burn bi-elliptic Mars orbit insertion
 - Apotwist
 - 3-burn bi-elliptic trans-Earth injection
- Setting up Mars parking orbit with potential constraints
 - Sub-periapse landing target
 - Lighting condition
 - Ascent from Mars surface

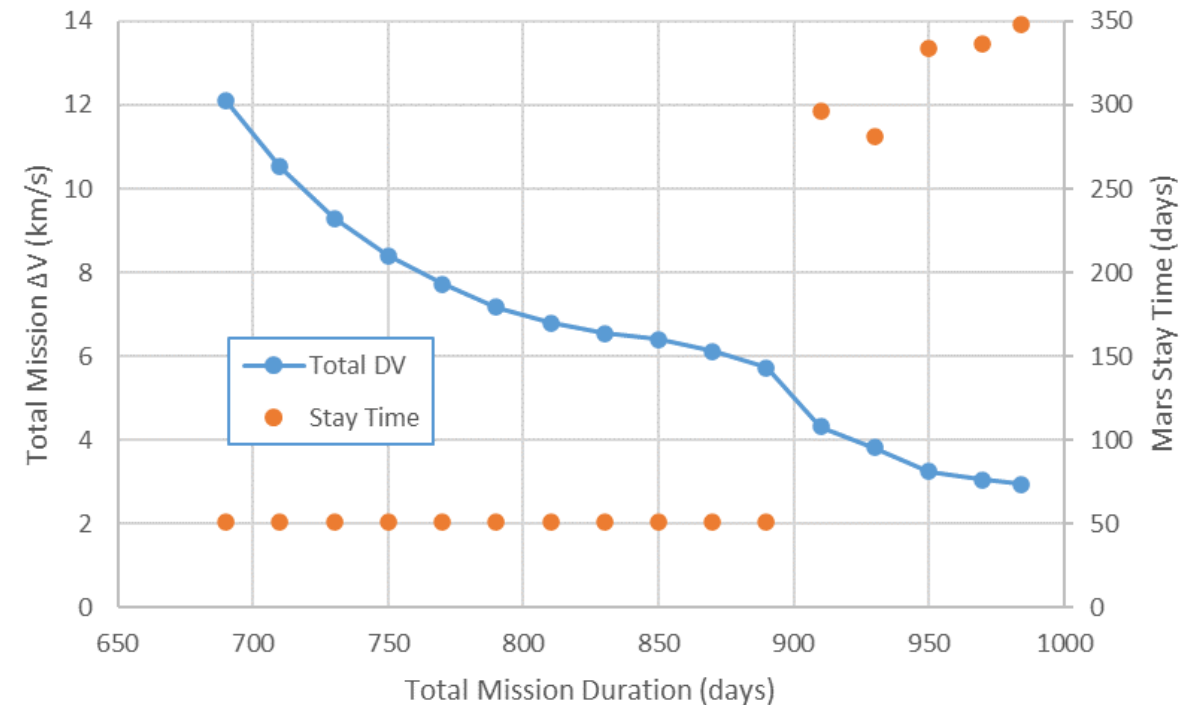


Roundtrip Duration Trades



- Modeling tool enables wide trade space analysis to understand impact of independent variables
 - Analysis Assumptions:
 - Depart Earth from 400x400,000 km orbit
 - 5-sol Mars parking orbit
 - 35 deg landing site latitude
 - Return to 400x400,000 km Earth orbit
- For current analysis to enable tool development & validation, developed model enables additional analysis of variation to assumptions.*
- SNOPT Optimization
 - 26 independent variables
 - 15 constraints

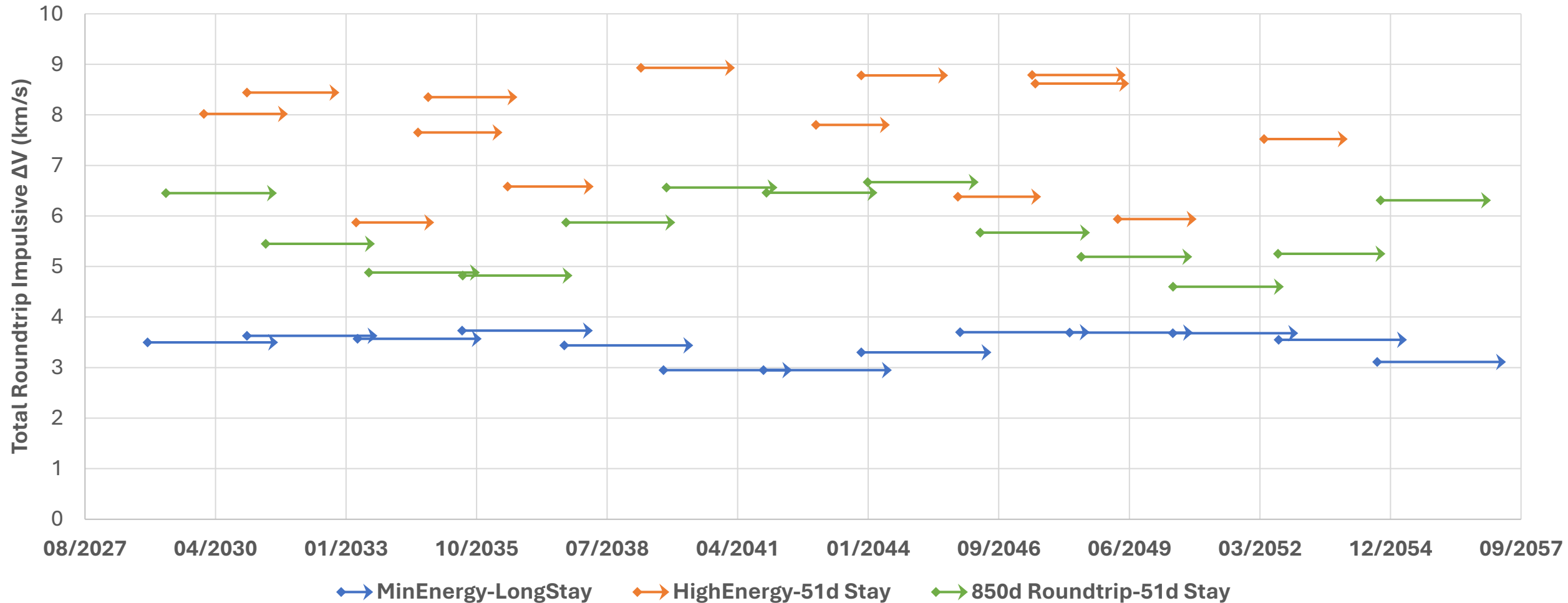
Example 2039 Mission Opportunity



Optimization Across Mission Opportunities



Optimized Roundtrip Mission ΔV Across Mission Opportunities, Line length Represents Roundtrip Duration



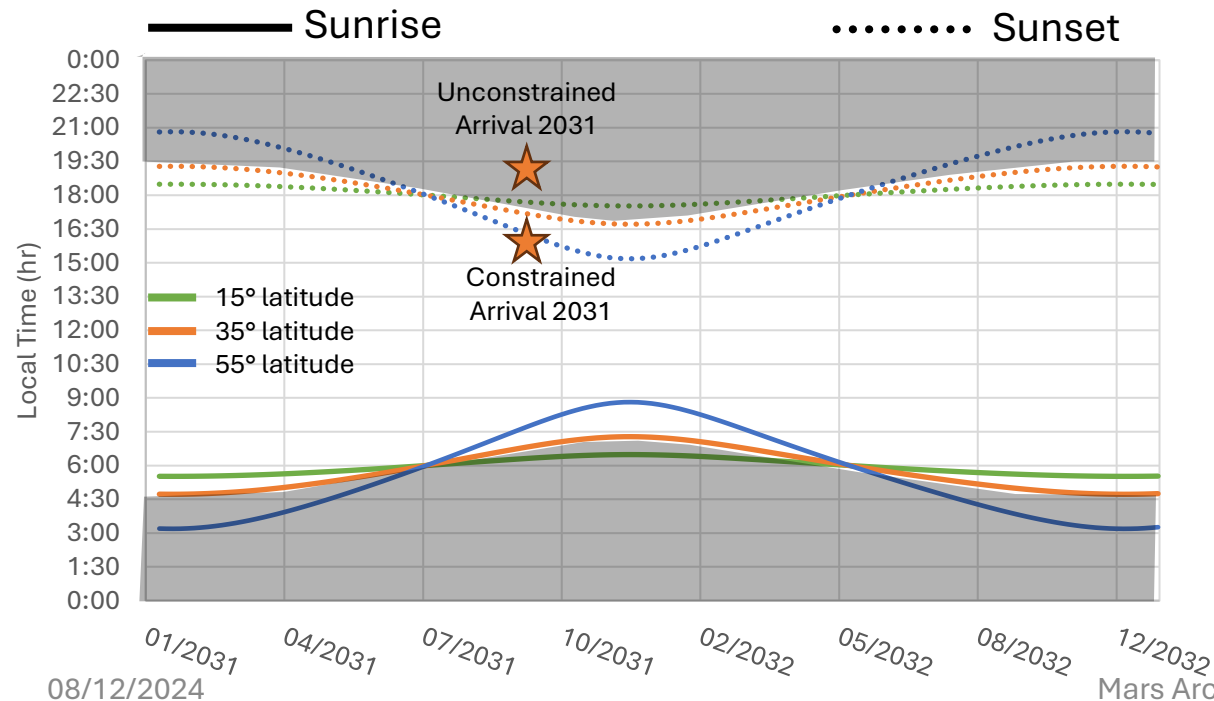
Enables Specialized Analysis – Lighting Condition at Landing



Mars parking orbit orientation determines the local time of landing regardless of the landing site longitude

(assuming direct, sub-periapse landing)

- Average daylight between 6:00 and 18:00 in Martian clock
- Sunrise and sunset times differ by latitudes and have seasonal variations



Options to improve lighting condition at landing:

1. Increase deorbit burn to target landing site short of periapse
 - Steeper flight path angle at entry, more difficult to abort
2. Loiter in parking orbit before descent
 - 10 minutes per 5-sol rev
3. Constrain the local time of landing in global optimization
 - Increase overall mission ΔV

Example 2031 Arrival 35° Latitude	Solution 0 Unconstrained	Solution1 Constrained for Local Landing between 0600-1800	Solution 2 Constrained for Local Landing between 0800-1600
TMI (km/s)	0.580	0.539	0.593
MOI (km/s)	1.399	1.479	1.726
ApoTwist (km/s)	0.012	0.022	0.014
TEI (km/s)	0.989	1.026	1.073
EOI (km/s)	0.654	0.652	0.646
Inclination APO (deg)	38.093	39.078	35.840
RAAN APO (deg)	76.071	61.143	41.659
Total ΔV (km/s)	3.634	3.718	4.052
Landing Local Time	19:06	17:51	15:57
Sunrise	06:51	06:58	06:52
Sunset	17:08	17:01	17:07

Summary

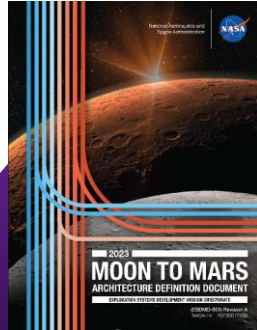


- Developed modeling capability to perform end-to-end trajectory simulation and optimization for high-thrust trajectories
- Bi-elliptic apotwist can be solved with globally optimized roundtrip trajectories while satisfying constraints on landing site, lighting condition, and Mars ascent
- Lighting condition is an important factor and must be considered in the mission design process
- ΔV is not the sole metric to compare different architecture options

Learn More about Human Spaceflight Architecture



nasa.gov/MoonToMarsArchitecture

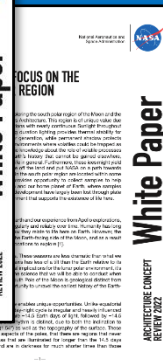
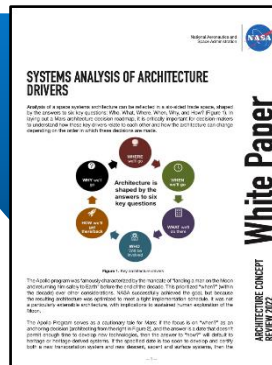


Architecture Definition Document

Detailed documentation of a snapshot of NASA's human spaceflight architecture and exploration strategy

Moon to Mars Architecture Summary

High-level overview of NASA's Moon to Mars architecture and exploration strategy



White Papers

19 papers on architecture study details for frequently discussed topics